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Lightweight QRS-attention network for efficient real-time detection of ECG anomalies

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Abstract
Real-time electrocardiogram (ECG) anomaly detection is essential for early diagnosis and continuous heart monitoring. However, deploying deep learning models in wearable or mobile environments is still limited by computational and memory constraints. In this paper, we introduce a lightweight QRS-attention network (QRSAN), optimized by quantization for efficient and low-latency arrhythmia detection. The proposed QRS-attention mechanism uses temporal convolution and causal attention to selectively highlight discriminative features within QRS complexes, enabling robust identification of abnormal heartbeats in streaming ECG signals. Experiments on the MIT-BIH Arrhythmia Database demonstrate that QRSAN achieves superior performance across multiple evaluation metrics, including accuracy, precision, recall, specificity, F1-score and latency. Compared with state-of-the-art lightweight baselines, QRSAN improves F1-score by up to twofold over LSTM and by approximately 2% over CNN–LSTM and EfficientNet1D, with comparable latency. These results indicate that QRSAN provides an effective and computationally efficient framework for continuous ECG anomaly monitoring in wearable and edge-computing systems.

Keywords Electrocardiogram, Attention mechanism, Deep learning

1 Introduction

Cardiovascular diseases remain the leading cause of death worldwide, surpassing all cancers and respiratory diseases combined, as shown in Fig. 1. This overwhelming global burden emphasizes the necessity of reliable cardiac monitoring systems that can operate continuously and accurately in daily life [1]. Among various physiological signals, Electrocardiography (ECG) is a non-invasive diagnostic tool that records the electrical signals of the heart and is widely used in the diagnosis of various cardiovascular diseases[2]. In particular, ECG analysis plays a crucial role in the early detection of cardiac abnormalities such as arrhythmia, ischemia, and atrial fibrillation[3, 4]. Precision and real-time ECG monitoring are especially important in wearable and mobile healthcare systems that are currently being actively researched and developed, enabling rapid medical response and early diagnosis outside the clinical setting[5]. However, implementing deep learning based ECG anomaly detection on resource-constrained platforms such

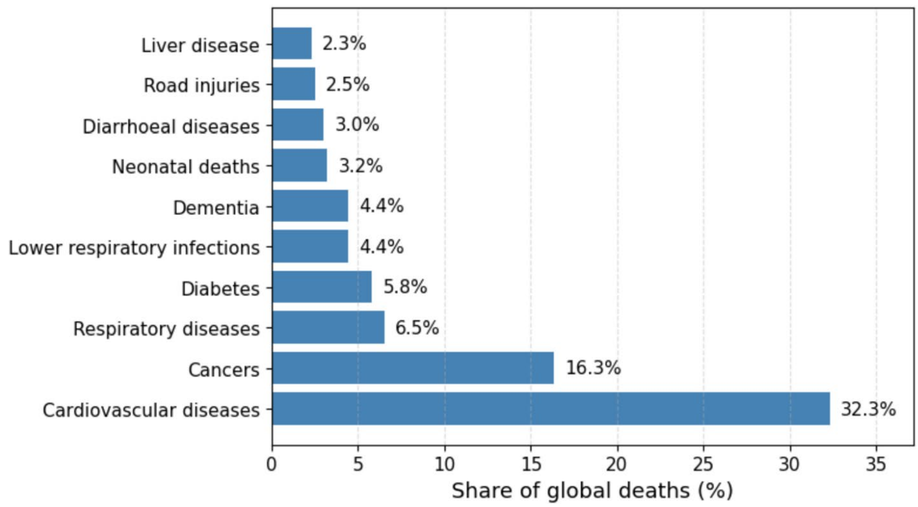


Fig. 1 Global distribution of major causes of death. Cardiovascular diseases account for more than one-third of all fatalities, underscoring the need for effective cardiac monitoring and early detection technologies

as wearables and mobiles poses significant challenges. Traditional convolutional neural networks or recurrent neural networks often require a large number of parameters, high computational complexity, and significant energy consumption, limiting their application in real-time scenarios and embedded hardware on ECG data[6].

To address these limitations, this study proposes a lightweight **QRS-Attention Network (QRSAN)** that incorporates a QRS-focused attention mechanism and quantization to achieve efficient and accurate real-time ECG anomaly detection.

QRS complexes representing ventricular depolarization in ECG waveforms, which will be described in subsequent sections, are of particular clinical importance as their morphology and timing directly indicate conduction and rhythm disorders[7]. Therefore, we focus on these QRS waveforms, aiming to efficiently detect abnormal heart events while maintaining interpretability tailored to physiological implications. The proposed QRS-attention mechanism uses causal time convolution for high accuracy to highlight characteristic patterns within QRS complexes.

The major contributions of this work are summarized as follows:

- *QRS-Focused Attention Mechanism:* A novel causal temporal attention structure is introduced to selectively amplify morphological and temporal variations within QRS complexes, enabling accurate anomaly detection under streaming ECG conditions.
- *Quantization-Based Optimization:* A data-level non-uniform ECG amplitude discretization scheme is introduced to reduce inference latency and memory access cost, without modifying network weights or architecture.

Comprehensive experiments conducted on the MIT-BIH arrhythmia database show that QRSAN achieves competitive diagnostic performance compared to existing methods. Through our proposed study, we provide practical and scalable solutions for real-time cardiac anomaly detection to address the limitations of resource-constrained mobile or wearable healthcare systems.

The rest of this paper is organized as follows: Sect. 2 covers the ECG, attention mechanisms, and background knowledge on quantization underlying this paper. Section 3 describes the entire system design and algorithms for the proposed QRSAN model.

Section 4 describes the experimental setup, evaluation metrics, and comparison results using the MIT-BIH arrhythmia database. Finally, Sect. 5 concludes the paper and outlines future research directions.

2 Related work

In this section, the three main areas related to the proposed research are as follows: (1) the medical significance of QRS-focused analysis in ECG interpretation, (2) the application of attention mechanisms for feature selection and interpretability, and (3) quantization techniques for lightweight deep neural network deployment.

2.1 Electrocardiogram and the significance of QRS complexes

ECG signals exhibit a series of repeating waves that correspond to distinct physiological events within the cardiac cycle. As illustrated in Fig. 2, the QRS complex is generated during *ventricular depolarization*—the instant the ventricles contract to pump blood—representing the "beat" component of the heart sound. The subsequent T wave reflects *ventricular repolarization*, corresponding to the recovery phase that precedes the next contraction, or the "pause" between beats. Abnormalities in these waveforms are strong clinical indicators of cardiac disease[8]. A widened or distorted QRS complex often reflects *myocardial infarction* or conduction block, whereas irregularities in the T wave suggest *ischemic heart disease* or repolarization defects[9, 10]. In addition, atypical R-peak morphology is associated with *arrhythmia* and asynchronous ventricular activation[11]. Therefore, accurate characterization of QRS and T-wave morphology is essential for reliable cardiac diagnosis.

From a data-dimensional perspective, ECG signals are typically represented as one-dimensional (1D) time-series data because they measure voltage changes over time to reflect dynamic heart behavior[12].

A standard clinical ECG employs twelve leads, each representing a different spatial projection of the electrical potential of heart. While this multi-lead configuration offers spatial comprehensiveness, it also introduces redundancy, inter-lead variability, and synchronization challenges, which complicate automated interpretation[13]. Therefore, it is crucial to develop robust, lead-invariant algorithms capable of extracting consistent and physiologically meaningful features from 1D ECG data.

As shown in Fig. 2, the QRS complex contains the highest local energy in the ECG signal, characterized by sharp, high-amplitude transients, making it a characteristic region

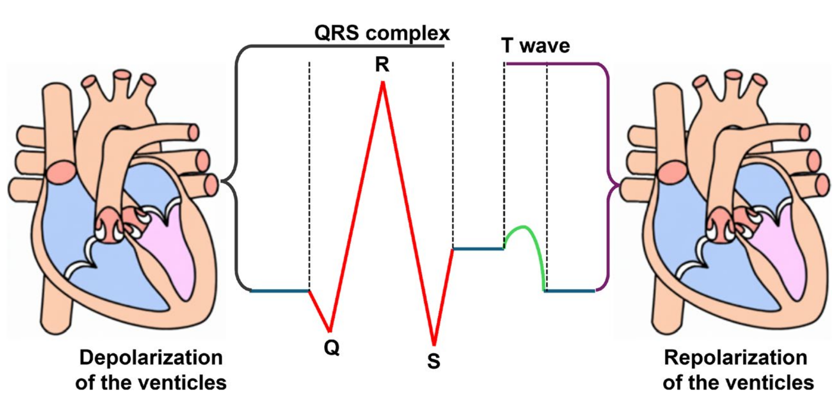


Fig. 2 The waveform of an electrocardiogram (ECG) signal. The QRS complex represents the ventricular depolarization phase, while the T wave indicates the ventricular repolarization phase leading to the end of the systolic cycle

for modeling and feature extraction[14]. We form the conceptual basis of the QRS-attention network of subsequent Sect. 3 from the perspective that focusing our analytical focus on these QRS regions allows our algorithm to leverage physiologically relevant and informative segments to improve both classification accuracy.

2.2 Attention mechanisms for precise networks

The attention mechanism underlying the recently widely used language model is a key concept in modern deep learning, allowing the model to dynamically focus on the most informative parts of the input sequence[15]. Unlike traditional convolutional neural network or recurrent neural network architectures, attention gains performance by assigning different importance weights to different input elements, capturing the long-range dependence and contextual relevance of the model[16, 17].

Basically, the attention mechanism is given an input sequence represented by a set of *n* feature vectors $\mathrm{X} = [\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_n]$, each element is projected into three latent spaces known as *queries*, *keys*, and *values*:

$$\mathrm{Q} = \mathrm{XW}_Q, \quad \mathrm{K} = \mathrm{XW}_K, \quad \mathrm{V} = \mathrm{XW}_V,$$

where $\mathrm{W}_Q, \mathrm{W}_K, \mathrm{W}_V$ denote learnable weight matrices. The attention output is computed as a weighted sum of the value vectors, where the weights are determined by the similarity between queries and keys:

$$\mathrm{Attention}(\mathrm{Q}, \mathrm{K}, \mathrm{V}) = \mathrm{softmax}\left(\frac{\mathrm{QK}^\top}{\sqrt{d_k}}\right)\mathrm{V},$$

with d_k being the key dimension used for scaling. This work enables deep learning models with attention mechanisms to adaptively pay attention to elements that are contextually relevant to the representation of the targeted data. As such, the attention framework becomes the theoretical basis for the proposed QRS-centric attention mechanism by enabling data-driven feature prioritization.

Recent studies have suggested appropriate attention mechanisms based on the features implied by various time series data, resulting in improved performance such as detection and analysis[17–20]. This approach dynamically weights the time segments or channels that contribute the most to anomaly detection, improving classification accuracy and interpretability. However, existing attention mechanisms, especially global self-attention mechanisms, are limited in their application to real-time systems due to their high computational costs[21]. To address these issues, the proposed QRS-attention mechanism aims to enable efficient and explainable anomaly detection under resource-constrained conditions using causal temporal convolution structures that explicitly focus on QRS complexes.

2.3 Quantization for lightweight neural networks

Quantization is one of the most effective techniques for reducing the computational and memory needs of deep neural networks, especially in real-time and edge computing applications[22]. The underlying principle is to map continuous value parameters or activations to finite sets of discrete levels, reducing numerical precision and overall storage requirements. In most implementations, this can be achieved through uniform quantization that divides the entire range of values of the signal or weight matrix by equal intervals[23]. These methods are popular in mobile and embedded inference systems for their simple hardware implementation and compatibility with traditional digital accelerators.

However, such an ideal uniform quantization assumes that the data distribution is evenly distributed over the quantization range, and this assumption is difficult to establish in practice. When abnormalities are extremely partial, with the majority of normal data, they often follow very nonlinear or heavy tail distributions, with many values clustered near zero, while some extremities dominate the dynamic range. Uniform quantization under these conditions results in uneven quantization errors, sacrificing precision in dense regions and assigning redundant precision to rarely occurring values. In biomedical signals such as ECG, this problem is amplified because the amplitude and energy distribution vary greatly depending on the patient and the heart condition. This makes stationary phase quantization suboptimal for diagnostic applications.

To address these limitations, recent work focuses on data adaptive quantization frameworks that dynamically adjust quantization boundaries based on data statistics or signal entropy. This aims to maintain feature discrimination while ensuring computational efficiency by allocating a finer quantization resolution to regions of high probability density to minimize information loss[24, 25].

The evolution to data-driven quantization establishes the conceptual basis of this work. In this work, we demonstrate the proposed performance improvement by applying quantization only at the data level to discretize ECG signal amplitudes, rather than quantization of network weights or activations.

3 System design

As shown in Fig. 3, the proposed **Lightweight QRS-Attention Network (QRSAN) detection model** consists of two core components:

- QRS-attention mechanism emphasizing the morphological and temporal features inherent in QRS signals.
- Quantization-based lightweight ECG anomaly detection model with QRS-attention mechanism.

3.1 QRS-attention mechanism

The proposed **QRS-Attention Mechanism** is a causal and computationally efficient module that selectively enhances morphological and temporal features associated with the QRS complex in ECG signals. The mechanism combines causal temporal convolutions, residual refinement, and dual-path attention scoring to emphasize ventricular depolarization patterns that are diagnostically relevant.

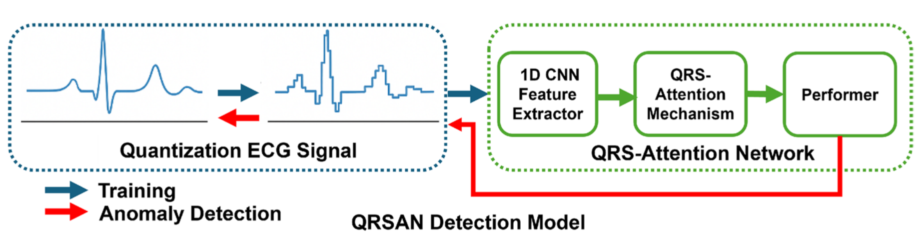


Fig. 3 The structure of the proposed QRSAN detection model

Let an input ECG sequence be represented as

$$\mathrm{X} = [x_1, x_2, \ldots, x_T] \in \mathbb{R}^{T\times 1},$$

where T is the number of temporal samples per segment. The model first applies a causal convolutional layer to extract local QRS morphology[26]:

$$\mathrm{Q} = \phi(\mathrm{X} * \mathrm{W}_q + \mathrm{b}_q), \quad \phi(\cdot) = \mathrm{ReLU},$$

where W_q is a convolution kernel of size 5, and causal padding ensures that Q_t depends only on $\{x_1, \ldots, x_t\}$.

To adaptively emphasize pathological depolarization patterns, a gating operation is introduced[6]. A convolutional gate produces a pathological activation map

$$\mathrm{g} = \sigma(\mathrm{Q} * \mathrm{W}_g + \mathrm{b}_g),$$

where $\sigma(\cdot)$ is the sigmoid function and W_g is a kernel of size 3. The gated representation is then defined as

$$\widetilde{\mathrm{Q}} = \mathrm{Q} \odot (1 + \alpha \mathrm{g}),$$

where $\alpha = 0.8$ in our implementation and $\odot$ denotes element-wise multiplication. This operation amplifies feature responses corresponding to morphological deviations in the QRS region while preserving causal flow[27].

The gated features $\widetilde{\mathrm{Q}}$ are then processed by two residual Temporal Convolutional Network (TCN) blocks with increasing dilation factors $d \in \{1, 2\}$. Each block applies two causal convolutions with dilation rates d and $2d$:

$$\mathrm{H}^{(d)} = \mathrm{LN}\big(\phi(\mathrm{W}_{d,2} * \phi(\mathrm{W}_{d,1} * \widetilde{\mathrm{Q}} + \mathrm{b}_{d,1}) + \mathrm{b}_{d,2})\big) + \mathrm{Proj}(\widetilde{\mathrm{Q}}),$$

where LN denotes layer normalization and Proj represents a 1×1 projection for residual alignment when channel dimensions differ. As a result, the temporal representation is

$$\mathrm{H} = \mathrm{H}^{(2)}\big(\mathrm{H}^{(1)}(\widetilde{\mathrm{Q}})\big).$$

To derive the attention weights, two parallel scoring paths estimate temporal and morphological salience:

$$a_t^{(\mathrm{temp})} = \sigma(\mathrm{w}_t^\top \mathrm{h}_t + b_t), \quad a_t^{(\mathrm{morph})} = \sigma(\mathrm{w}_m^\top \widetilde{\mathrm{q}}_t + b_m),$$

where h_t and $\widetilde{\mathrm{q}}_t$ denote the temporal and morphological feature vectors at time t. The fused attention score is obtained by multiplicative interaction:

$$s_t = a_t^{(\mathrm{temp})} \cdot a_t^{(\mathrm{morph})},$$

followed by a time-wise normalization using a numerically stable softmax:

$$A_t = \frac{\exp(s_t - s_{\max})}{\sum_{j=1}^{T} \exp(s_j - s_{\max})}, \quad s_{\max} = \max_j s_j.$$

This normalization enforces positive, unit-sum attention weights while preventing overflow during exponential computation. The final attention-weighted output is given by

$$y_t = x_t \cdot A_t, \qquad \mathrm{Y} = [y_1, y_2, \ldots, y_T].$$

In terms of computational complexity, all operations scale linearly with input length T,

$$\mathcal{O}(T \cdot F \cdot k),$$

where F is the feature dimension and k the convolution kernel size, significantly more efficient than global self-attention with $\mathcal{O}(T^2)$ complexity. The QRS-Attention mechanism aims to combine time-dependent modeling with morphology-aware gating to provide a lightweight, interpretable, and physiologically reasonable approach to real-time ECG analysis.

Algorithm 1 Quantization-based Lightweight ECG Anomaly Detection with QRS-Attention

Require: Raw ECG segment $x \in \mathbb{R}^{1\times T}$, quantization levels L, pretrained centroids $\{\mu_\ell\}_{\ell=1}^{L}$
Ensure: Predicted label $\hat{y}$
Data Quantization
for $t = 1$ to T **do**
$\quad \widehat{x}[t] \Leftarrow \mu_{\arg\min_\ell \|x[t]-\mu_\ell\|^2}$ ▷ K-Means–based non-uniform quantization
end for
1D CNN Feature Extraction
$\mathbf{F} \Leftarrow \mathrm{MaxPool1D}\big(\mathrm{ReLU}(\mathrm{Conv1D}_{5\times 32}(\mathrm{ReLU}(\mathrm{Conv1D}_{7\times 1}(\widehat{x}))))\big)$
QRS-Attention Encoding (Causal TCN)
$\mathbf{Q} \Leftarrow \mathrm{ReLU}(\mathrm{Conv1D}^{\mathrm{causal}}_{5\times 64}(\mathbf{F}))$
$\mathbf{g} \Leftarrow \sigma(\mathrm{Conv1D}^{\mathrm{causal}}_{3\times 1}(\mathbf{Q}))$
$\widetilde{\mathbf{Q}} \Leftarrow \mathbf{Q} \odot (1 + 0.8\mathbf{g})$
for $d \in \{1, 2\}$ **do**
$\quad \mathbf{U} \Leftarrow \mathrm{ReLU}(\mathrm{Conv1D}^{\mathrm{causal}}_{3,d}(\widetilde{\mathbf{Q}}))$
$\quad \widetilde{\mathbf{Q}} \Leftarrow \mathrm{ResidualAdd}(\widetilde{\mathbf{Q}}, \mathrm{LayerNorm}(\mathbf{U}))$
end for
$\mathbf{S} \Leftarrow \mathrm{time\_softmax}\big(\sigma(\mathrm{Conv1D}_{1\times 1}(\widetilde{\mathbf{Q}})) \odot \sigma(\mathrm{Conv1D}_{1\times 1}(\mathbf{Q}))\big)$
$\mathbf{F}_{\mathrm{att}} \Leftarrow \mathbf{F} \odot \mathbf{S}$
Performer Classification
$\mathbf{H} \Leftarrow \mathrm{Performer}(\mathbf{F}_{\mathrm{att}})$
$\mathbf{z} \Leftarrow \mathrm{MLP}(\mathrm{GlobalAvgPool1D}(\mathbf{H}))$
$\hat{y} \Leftarrow \arg\max_c z_c$

3.2 QRSAN model

The proposed QRSAN model combines a distribution-aware data quantization scheme with QRS-Attention Mechanism to achieve both diagnostic accuracy and computational efficiency. This detection framework is designed to operate in three stages: convolutional feature extraction, QRS-Attention Mechanism, and Performer-based classification.

The input ECG signal $X \in \mathbb{R}^{1\times T}$ is first processed by a one-dimensional convolutional front-end that extracts waveform morphology while suppressing noise and baseline drift. Two convolutional layers with kernel sizes of 7 and 5, followed by ReLU activation and a max-pooling operation, progressively encode local temporal patterns corresponding to

P–QRS–T segments. This stage produces a compact spectral–temporal representation that retains key morphological characteristics necessary for arrhythmia classification.

Subsequently, temporal context and depolarization-related information are modeled through a dilated Temporal Convolutional Network (TCN) equipped with the QRS-Attention mechanism described in Sect. 3.1. The TCN employs causal convolutions to maintain real-time inference capability, while the attention module adaptively enhances feature responses around the QRS complex using a gating function that emphasizes pathological deviations in ventricular depolarization. Through this mechanism, the network selectively amplifies the most informative temporal intervals, reducing redundancy and improving interpretability of the learned feature space.

The advanced features are then delivered to a recently researched Performer-based lightweight classifier that is appropriate for ECG data[28]. Unlike conventional self-attention, the Performer approximates the quadratic attention operation using kernel-based linearization[29]. This allows efficient modeling of temporal dependencies across multiple cardiac cycles while maintaining feasibility for embedded deployment.

Prior to model training, the input ECG data are subjected to non-uniform quantization based on K-Means clustering. Given the training samples $\{x_i\}_{i=1}^{N}$, the global amplitude distribution is partitioned into L clusters by minimizing intra-cluster variance:

$$\underset{\{\mu_l\}_{l=1}^{L}}{\operatorname{argmin}} \sum_{i=1}^{N} \min_{1 \le l \le L} \|x_i - \mu_l\|^2,$$

where μ_l denotes the centroid of cluster l. Each signal sample x_i is then mapped to its nearest centroid $\widehat{x}_i = \mu_{l^*}$, defining a quantization level set $\mathcal{Q} = \{\mu_1, \ldots, \mu_L\}$ for $L \in \{4, 8, 16\}$. This process captures the natural distribution of ECG amplitudes, particularly preserving high-energy regions around QRS peaks while reducing granularity in low-variance intervals. During inference, the same centroid of cluster l obtained from the training phase are reused, ensuring distributional consistency between training and deployment without additional calibration overhead.

The quantized signal $\widehat{X}$ is subsequently processed by the convolutional–attention–transformer pipeline described above:

$$\mathrm{y} = f_{\text{Performer}}\big(f_{\text{QRS-ATT}}(f_{\text{CNN}}(\widehat{X}))\big),$$

where $f_{\text{CNN}}(\cdot)$ represents the convolutional encoder, $f_{\text{QRS-ATT}}(\cdot)$ denotes the temporal QRS-Attention network, and $f_{\text{Performer}}(\cdot)$ is the final classifier.

This entire process is detailed in the Algorithm. 1, and the integration of data-level quantization and domain-aware attention architecture implements a compact, hardware-efficient ECG anomaly detection model suitable for real-time arrhythmia inspection on edge or wearable devices.

4 Experiments

This section describes the experimental setup, evaluation metrics, and comparative results that validate the performance of the proposed QRSAN detection model. All experiments were conducted to assess both diagnostic accuracy and computational efficiency.

4.1 Experimental environment and dataset

The experiments were conducted using a **MIT-BIH Arrhythmia Database** configured as shown in Fig. 4. Because the dataset does not provide patient-level identifiers due to privacy-preserving protocols, all experiments were conducted using ECG beat level segmentation. The ECG classes within the dataset are classified into five clinical classes and labeled from 0 to 4 for the convenience of in this experiment: *Normal Beats*, *Supraventricular Ectopy Beats*, *Ventricular Ectopy Beats*, *Fusion Beats*, and *Unclassifiable Beats*[30, 31]. The full dataset contains 109,446 labeled beats, occupying 158.7 MB of memory. The predefined training and test sets provided by the dataset were used, and a stratified random split was applied to the training set to construct a validation set while preserving class distribution. Each ECG segment consists of 500 samples directly extracted from the provided waveform without additional detection or rearrangement.

The representative waveform of each ECG class is shown in Fig. 5, and the morphological differences based on the QRS and T-wave patterns in which these samples are seen are critical in distinguishing between normal and abnormal states.

Signal preprocessing techniques such as normalization, filtering, or data augmentation are not applied, and non-uniform K-means clustering was performed using only training data for quantized data experiments and applied to validation and test sets.

To ensure consistent evaluation, the proposed QRSAN-based lightweight detection model and all comparison models were evaluated under the same experimental conditions as shown in Table 1.

The hyperparameters of the proposed QRSAN model were selected to ensure stable training and fair comparison across models. The AdamW optimizer was employed together with weight decay regularization to balance convergence speed and generalization performance. To mitigate overfitting, early stopping was applied, and the learning rate was adaptively adjusted using the ReduceLROnPlateau scheduler. All experiments were conducted using fixed random seeds to ensure reproducibility. These hyperparameter settings were kept consistent across all baseline and proposed models for fair performance comparison.

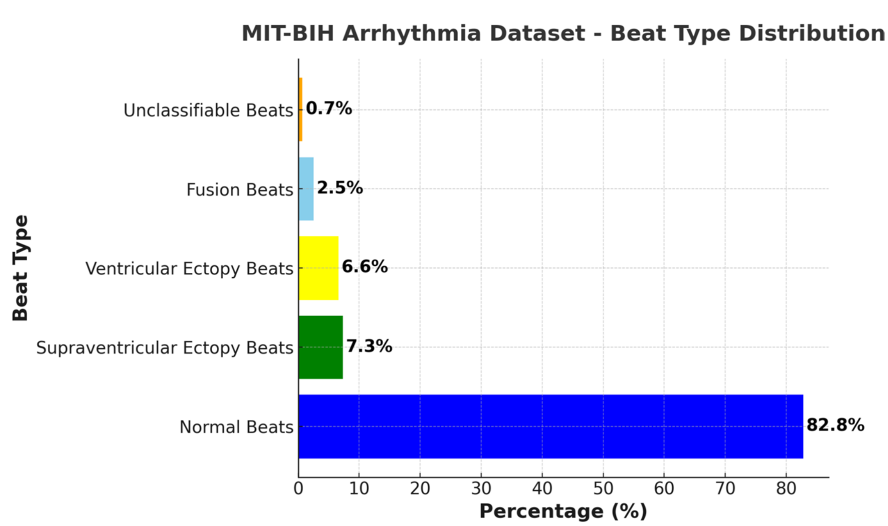


Fig. 4 Distribution of beat types in the MIT-BIH arrhythmia database

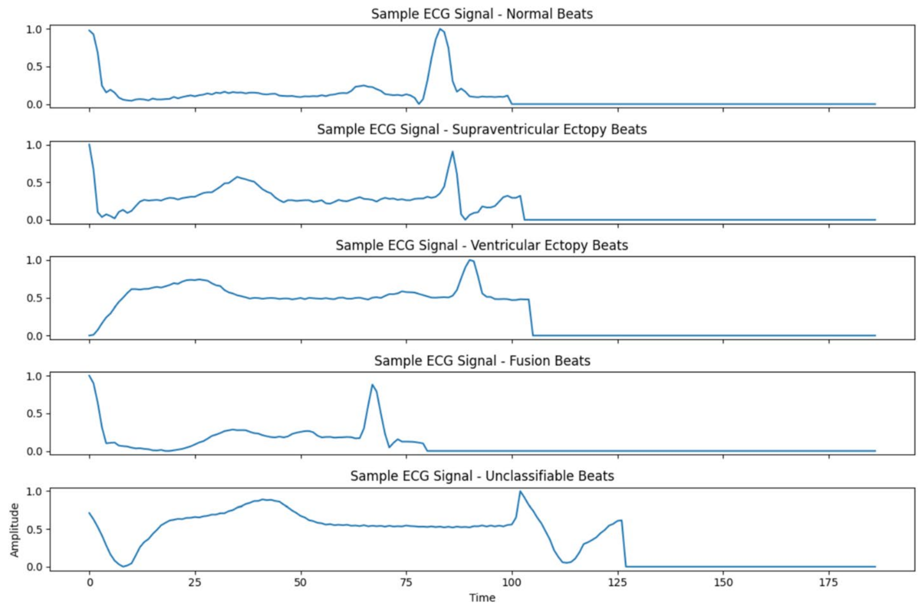


Fig. 5 Representative ECG waveforms of the five beat categories used in this study

Table 1 Experimental environment and hyperparameter settings

Parameter	Configuration
Hardware environment	Intel Core i9-12900K CPU, 64 GB RAM, NVIDIA RTX 3090 (24 GB)
Batch size	512
Learning rate	1×10^{-3}
Epochs	10 (early stopping applied to prevent overfitting)
Weight decay	1×10^{-4}
Optimizer	AdamW
Scheduler	ReduceLROnPlateau (factor = 0.5, patience = 3)
Loss function	CrossEntropyLoss

4.2 Evaluation metrics

Performance evaluation consisted of two complementary aspects: model efficiency and diagnostic accuracy.

4.2.1 Efficiency metrics

To quantify the efficiency of each network, a given model M and an input ECG segment $x \in \mathbb{R}^{1\times1\times500}$, latency τ in milliseconds per inference was computed as:

$$\tau = \frac{1}{N}\sum_{i=1}^{N}\left(t_i^{(\mathrm{end})} - t_i^{(\mathrm{start})}\right) \times 1000,$$

where N denotes the number of repeated inferences, and GPU synchronization was applied before and after each iteration to eliminate asynchronous timing artifacts. This latency measure reflects the real-time suitability of the model for deployment on edge devices, particularly emphasizing the computational benefits achieved through quantization and lightweight architecture design. All latency measurements in this study were conducted in a controlled benchmarking environment on the NVIDIA RTX 3090 GPU platform to ensure fair and reproducible comparisons between models. While no direct deployment experiments have been conducted on wearable devices, future studies will focus on validating the proposed framework on actual wearable or embedded hardware platforms.

4.2.2 Diagnostic metrics

To evaluate detection accuracy for normal and abnormal beats, the following six indices were used:

$$\begin{aligned}
\text{Accuracy} &= \frac{TP+TN}{TP+TN+FP+FN},\\
\text{Precision} &= \frac{TP}{TP+FP},\\
\text{Recall} &= \frac{TP}{TP+FN},\\
\text{Specificity} &= \frac{TN}{TN+FP},\\
\text{F1-Score} &= \frac{2\,(\text{Precision}\cdot\text{Recall})}{\text{Precision}+\text{Recall}},\\
\text{AUC-ROC} &= \int_0^1 TPR(FPR)\,d(FPR),
\end{aligned}$$

where *TP*, *TN*, *FP*, and *FN* denote true positives, true negatives, false positives, and false negatives, respectively. Higher values of all metrics correspond to better diagnostic performance[32].

4.3 Baseline models for comparison

To validate the effectiveness of the proposed QRSAN model, five representative lightweight and high-performance ECG classification models were selected for comparison. These baselines cover a broad spectrum of architectural paradigms, including convolutional, recurrent, hybrid, and transformer-based designs widely adopted in recent ECG anomaly detection research.

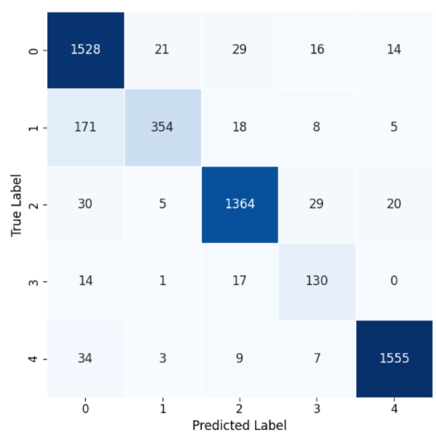
- *1D-ResNet* [33]: Proposed by Rajpurkar et al., this model employs a 34-layer one-dimensional residual neural network that achieves cardiologist-level arrhythmia detection accuracy. The residual connections facilitate stable gradient propagation and deep feature extraction, making it a strong benchmark for ECG classification.
- *1D-EfficientNet* [34]: Based on the EfficientNet architecture by Tan et al., this model utilizes compound scaling to balance depth, width, and resolution, achieving high performance with significantly fewer parameters compared to conventional ResNet variants. Its efficiency makes it well suited for lightweight biomedical applications.
- *LSTM* [35]: Following Saadatnejad et al., this recurrent model captures long-term temporal dependencies using sequential memory gates without convolutional preprocessing. It demonstrates that arrhythmia classification can be achieved effectively from pure temporal representations of ECG signals.
- *CNN-LSTM* [36]: As introduced by Alamatsaz et al., this hybrid network combines a convolutional front-end for morphological feature extraction with a recurrent back-end for temporal context learning. The architecture enhances both local pattern recognition and long-term temporal modeling in ECG signals, offering an interpretable balance between accuracy and complexity.

- *1D Transformer Encoder* [37]: Proposed by Meng et al., this lightweight transformer variant applies multi-head self-attention to 1D ECG sequences, enabling the model to capture global temporal dependencies. Despite its superior modeling capacity, transformer-based ECG models typically involve higher computational cost and are thus useful baselines to evaluate the efficiency gains of QRSAN.

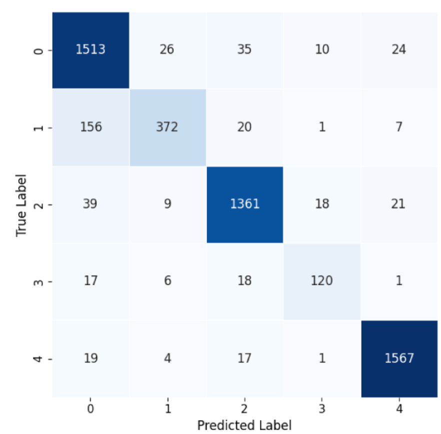
We used representative lightweight and attention-based ECG classification models published in the last five years for comparison, covering convolutional, recurrent, hybrid, and transformer-based paradigms. All comparison models were implemented under the same pre-processing and evaluation conditions to ensure fair comparison. This configuration allows a consistent analysis of diagnostic accuracy and latency performance relative to the proposed model.

4.4 Performance evaluation

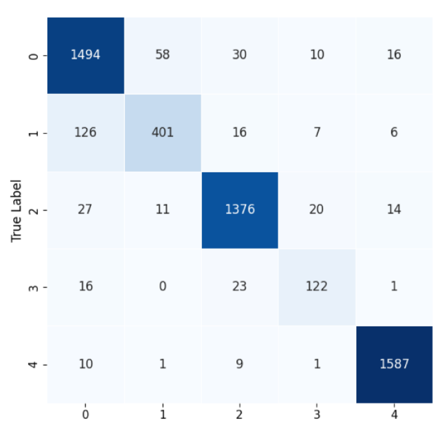
The initial experiment was conducted using the baseline Performer without QRSAN. The confusion matrix in Fig. 6a confirms that most misclassifications occur between



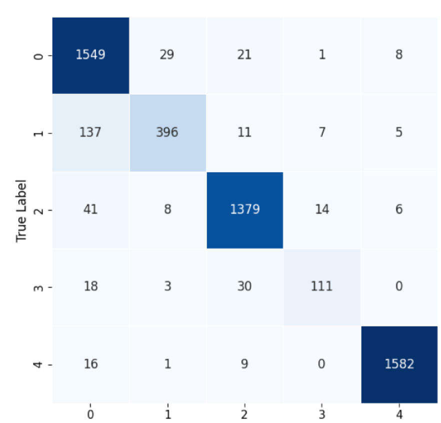
(a) Confusion matrix and performance metrics of the Performer baseline.



(b) Confusion matrix and metrics for QRSAN (4-bit quantization).



(c) Confusion matrix and metrics for QRSAN (8-bit quantization).



(d) Confusion matrix and metrics for QRSAN (16-bit quantization).

Fig. 6 Confusion matrices comparing performance of the baseline and the proposed QRSAN models with different quantization levels morphologically similar beat types, such as supraventricular (1) and ventricular ectopic beats (2). Overall diagnostic performance achieved an accuracy of 0.916, F1-score of 0.862, and AUC of 0.98, with an average inference latency of 12.4 ms per segment on the RTX 3090 platform.

Quantized QRS-Attention Network (QRSAN).

To further enhance efficiency, the Performer backbone was extended with the proposed QRS-Attention mechanism and quantized using the K-Means–based non-uniform quantization described in Sect. 3.2. Three quantization levels—4-bit, 8-bit, and 16-bit—were evaluated while maintaining identical preprocessing and training conditions. Each configuration was assessed using the metrics defined in Sect. 4.2, with results summarized in Table 2 and confusion matrices illustrated in Fig. 6b–d.

At the 4-bit level, QRSAN achieved an accuracy of 0.9166 and F1-score of 0.8693 with an inference latency of 12.9 ms. Despite its coarse quantization, diagnostic performance remained comparable to the floating-point baseline, demonstrating robustness against low-bit precision. Moving to 8-bit precision yielded a balanced improvement—accuracy 0.9253, F1-score 0.8759, and AUC 0.9864. Finally, the 16-bit model slightly enhanced diagnostic quality-accuracy 0.9322, precision 0.9119, F1 0.8815, AUC 0.9855, while maintaining similar latency 13.5 ms.

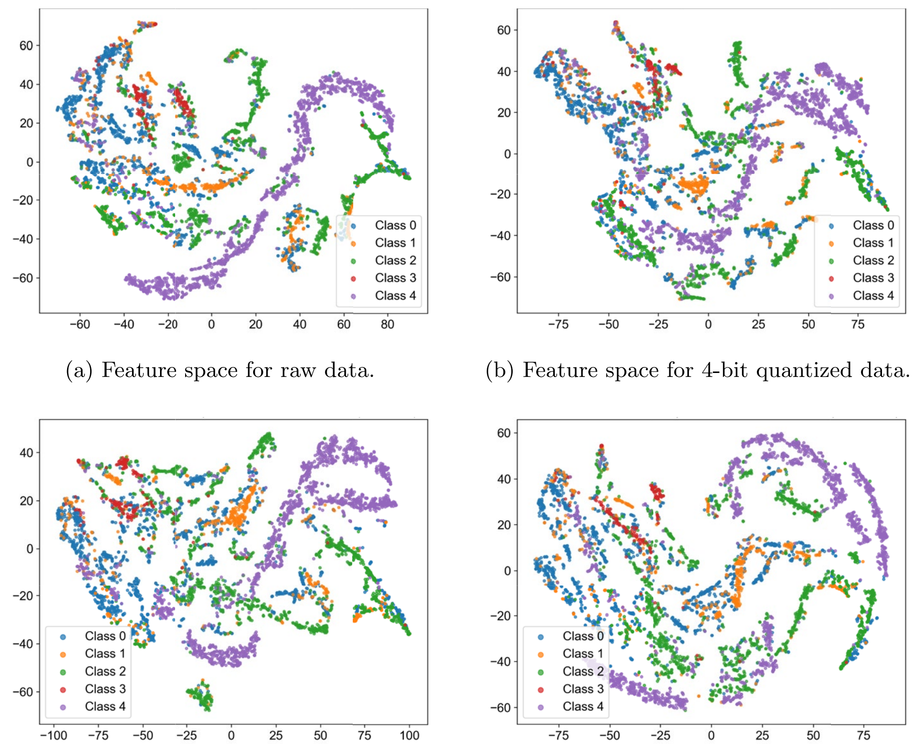
As shown in Fig. 3, the proposed QRSAN contains 223,176 trainable parameters. This parameter count is comparable to that of the Performer baseline model with 219,205 trainable parameters, confirming that the proposed QRS-aware attention network improves diagnostic performance and inference efficiency without introducing additional parameter overhead.

Among all configurations, the 8-bit model provides the optimal trade-off between diagnostic accuracy and computational cost, whereas the 16-bit version delivers marginally higher clinical precision at the expense of minimal latency increase.

We performed t-SNE visualization on latent features extracted from QRSAN to further analyze the effect of quantization on the feature representation. Fig. 7 show the distribution of the respective features obtained from the raw ECG signals and quantized inputs at the 4-bit, 8-bit, and 16-bit levels. As shown in Fig. 7a, the interclass boundaries for raw ECG data indicate the presence of an overall separation structure, particularly for Class 4. As shown in Fig. 7b, the application of 4-bit quantization increases the local mixing and the feature space is slightly more distributed, but the overall class-specific structure remains the same and the meaningful representation is maintained even at low levels of quantization. As shown in Fig. 7c, the 8-bit quantization model demonstrates more compact and stable clusters, suggesting that moderate quantization can regularize feature representations while preserving discriminative characteristics. As shown in Fig. 7d, the 16-bit quantization configuration provides a feature distribution similar to that of the raw input, but some classes exhibit a slight spread. Considered with Table 2,

Table 2 Performance comparison of quantized QRS-attention networks. Bold indicates the best performance per column.

Model	Accuracy	Precision	Recall	F1-score	AUC	Latency (ms)
Baseline performer	0.9162	0.8781	0.8597	0.8623	0.9800	12.39
QRSAN (4-bit)	0.9166	0.8930	0.8530	0.8693	0.9810	12.91
QRSAN (8-bit)	0.9253	0.8861	**0.8681**	0.8759	**0.9864**	14.10
QRSAN (16-bit)	**0.9322**	**0.9119**	0.8594	**0.8815**	0.9855	13.53



(a) Feature space for raw data.

(b) Feature space for 4-bit quantized data.

(c) Feature space for 8-bit quantized data.

(d) Feature space for 16-bit quantized data.

Fig. 7 Feature space analyzed by t-SNE for each quantization level

8-bit provides the most favorable trade-off as a regularized structure compared to raw input, suggesting that 16-bit is stable but additional gains are limited. These observations are consistent with quantitative results and confirm that the proposed QRSAN maintains a stable and interpretable representation, especially at the quantization level of 8-bit and 16-bit configurations. Overall, the feature space learned in all quantization settings maintains a clear class-specific clustering pattern, indicating that the proposed quantization scheme does not distort the discriminative representation.

This prominent 8-bit and 16-bit QRSAN detection model undergoes comparative performance evaluation against the baseline architectures introduced in Sect. 4.3 in the following sections.

Instead of compressing network parameters or reducing model storage sizes, the proposed non-uniform quantization simplifies the input signal representation, contributing to stable and efficient inference without modifying the model architecture or the number of parameters.

It is important that the goal of the proposed quantization-aware QRSAN framework is not to explore absolute latency minimization in certain bit settings, but rather to enable consistently robust and efficient inference in different quantization settings. While implementation-level overheads can lead to some lag time differences between 8-bit and 16-bit configurations in GPU environments, all quantized QRSAN variants show superior diagnostic performance and reliable inference efficiency compared to baseline models. From a system-level perspective, both 8-bit and 16-bit configurations represent valid operating points of the proposed QRSAN framework, which highlights flexibility and robustness under various precision constraints.

Table 3 Performance comparison of baseline ECG classification models and the proposed QRSAN. Bold indicates the best performance per column.

Model	Accuracy	Precision	Recall	F1-Score	AUC	Latency (ms)
CNN–LSTM	0.9201	0.8936	0.8468	0.8668	0.9851	8.08
EfficientNet1D	0.9127	0.8419	0.8864	0.8601	0.9847	16.67
ResNet1D	0.8218	0.8715	0.6284	0.6518	0.9672	11.56
Transformer1D	0.7549	0.6977	0.6525	0.6702	0.9217	13.77
LSTM	0.5362	0.4146	0.3769	0.3595	0.7853	8.68
QRSAN (8-bit)	0.9253	0.8861	**0.8681**	0.8759	**0.9864**	14.10
QRSAN (16-bit)	**0.9322**	**0.9119**	0.8594	**0.8815**	0.9855	13.53

4.5 Performance evaluation with comparative models

To validate the effectiveness of the proposed QRSAN, five representative baseline architectures were implemented under identical preprocessing and training conditions introduced in Sect. 4.3: CNN–LSTM, EfficientNet1D, ResNet1D, Transformer1D, and LSTM (Table 3).

Among the baseline models, CNN–LSTM achieved the highest F1-score as 0.8668, followed closely by EfficientNet1D as 0.8601, confirming that convolutional encoders effectively capture local ECG morphology when combined with temporal modeling. ResNet1D exhibited moderate accuracy as 0.8218, validating the benefit of residual learning but revealing limitations in modeling long-term temporal dependencies. In contrast, Transformer1D and LSTM achieved substantially lower F1-scores as 0.6702 and 0.3595, respectively.

Although certain models such as LSTM and Transformer1D achieved relatively short inference times (8–13 ms), their diagnostic performance was markedly inferior, indicating that computational efficiency alone cannot compensate for insufficient morphological understanding. The poor performance of LSTM further suggests that purely sequential architectures relying solely on temporal order struggle to represent complex ECG waveforms, where robust local feature extraction—typically provided by convolutional encoders—is essential. Similarly, the Transformer and ResNet1D baselines highlight that global attention or purely residual convolution, without localized physiological priors, is suboptimal for ECG morphology modeling and yields weak inter-class separability.

Compared with all baselines, the 8-bit and 16-bit QRSAN models demonstrate both superior diagnostic capability and efficient latency. Despite additional quantization layers, QRSAN surpasses CNN–LSTM by approximately 1.5–2 percentage points in F1-score while maintaining comparable runtime. These results emphasize that the proposed quantization-aware, QRS attention mechanism achieves an optimal balance between speed, accuracy, and interpretability for lightweight ECG anomaly detection. From a clinical deployment perspective, the improved class-wise discrimination and reduced inference latency observed in the proposed QRSAN framework are particularly beneficial for continuous ECG monitoring scenarios, where timely detection of abnormal cardiac rhythms is important for early clinical intervention.

Comparative models such as CNN-LSTM lower latency, but at the cost of significant degradation in diagnostic performance. On the other hand, the proposed QRSAN framework provides more reliable detection while maintaining competitive inference time, making it more suitable for real-world clinical monitoring scenarios.

5 Conclusion

In this work, we present a QRSAN detection model designed for lightweight and real-time ECG anomaly detection. The proposed model achieves superior diagnostic accuracy and low-latency performance compared with state-of-the-art approaches by incorporating QRS form–centric attention, which leverages the inherent morphology of ECG signals, together with efficient quantization techniques. In particular, in comparative evaluations, QRSAN achieved up to a **twofold improvement in F1-score over LSTM** and an **approximately 2% performance gain over CNN–LSTM and EfficientNet1D**, while maintaining similar or lower inference latency. These results confirm that domain-specific attention emphasizing physiological landmarks can substantially enhance both interpretability and computational efficiency in ECG-based diagnostic modeling.

While the proposed method in this paper was evaluated on the MIT-BIH Arrhythmia Database, a public benchmark widely used in ECG arrhythmia classification research, validation on diverse real-world device data and multi-population datasets will be an important direction for future work to further assess generalizability across various clinical settings. Future work will also explore optimization strategies aimed at further reducing inference latency without compromising diagnostic precision. Furthermore, extending the proposed QRS-centric attention paradigm to language models may contribute to a unified framework for explainable and edge-intelligent monitoring.

Acknowledgements
This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Climate, Energy & Environment (MCEE) of the Republic of Korea (No. RS-2022-KP002860) and also supported by the MSIT (Ministry of Science and ICT), Korea, under the ITRC (Information Technology Research Center) support program (IITP-2025-RS-2021-II211835) supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation).

Author contributions
Namkyung Yoon conceived the main research idea and drafted the manuscript. Sanghong Kim implemented the proposed model and conducted the experiments. Seunghyun Yoo performed literature review and contributed to the experimental analysis. Hwangnam Kim supervised the overall research, reviewed and refined the manuscript, and secured the research funding. All authors contributed to the discussion of results and approved the final version of the manuscript.

Funding
This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Climate, Energy & Environment (MCEE) of the Republic of Korea (No. RS-2022-KP002860) and also supported by the MSIT (Ministry of Science and ICT), Korea, under the ITRC (Information Technology Research Center) support program (IITP-2025-RS-2021-II211835) supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation).

Data availability
The experimental data used in this study are derived from the publicly available MITBIH Arrhythmia Database, which is hosted by the PhysioNet repository (https://www.physionet.org/content/mitdb/1.0.0/). The dataset was originally introduced by Moody and Mark [30].

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no Conflict of interest.

Received: 31 October 2025 / Accepted: 3 February 2026
Published online: 11 February 2026

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